

ECO209 Writing Assignment 1

Fall 2024

Professor Yip

Due date: Dec 4th, 2024

Background

In recent years, Canada has faced persistent challenges with productivity growth, a key factor for improving living standards and maintaining global competitiveness. According to Statistics Canada, labor productivity in the business sector declined by 1.8% in 2023, marking the third consecutive annual decrease. In a recent speech, Senior Deputy Governor Carolyn Rogers emphasized the importance of productivity growth for Canada, which sparked a widespread discussion about Canadian productivity. The goal of this assignment is to investigate the recent trend of Canadian productivity.

Instructions

You are a junior economist for a large Canadian company tasked with writing a background report on Canada's productivity growth since 2010. This report is divided into two parts:

- 1) Summarize the Bank of Canada's report on productivity
- 2) Analyze Canada's TFP growth and compare the growth to an assigned country.

Your report will go to the upper-level business executives within this company, all of whom have a moderate understanding of intermediate macroeconomic theory. You may assume this audience has taken ECO209 in the past and can follow descriptions of our economic model. You may reference model terms and variables *as long as* you properly define them. For example, **do not** simply say 'this will cause an increase in the Marginal Product of Labour'. Instead, say 'this will cause an increase in the Marginal Product of Labour (MPL), which tells how much additional output when a firm increases its labour input'. Wherever possible, use English terms to refer to model terms rather than mathematical terms/formulas.

Your report must have a **minimum** of one data plot. All plots must contain an accurate title, a legend for every data series, and labels on all axes. You may include the plots either in the body of the text or in an appendix at the end of the report. Ensure each plot is labelled with a convenient name for referencing in the text, i.e., 'Figure 1'. Extra plots can be included if it assists you in your argument in the essay.

Your company has strict formatting rules for this report that you must follow. The report must:

- Include a **brief** introduction that should provide a broad background for Canada's productivity slowdown and give an overview of your analysis of TFP.
- Be **typed, double-spaced**, and have a **font size of 12 points**.
- Be between **700-1000 words long**. This is a strict cut-off, and you will lose points for writing more than 1000 words. Put the word count of your assignment in parentheses at the end, like "(943 words)."
- **Include a header** (the first four lines) with your full name and student ID number, the course information (ECO209, 2024–2025), and the title of your assignment.
- **Cite the data sources and required background** text from the Bank of Canada. You do not need to cite the lectures, class notes, or the textbook. However, if you use any other

articles (including the optional background paper) you must cite them. All references should be listed at the end of the paper. This reference list does not count towards the final word limit.

Assignment Details

Part 0: Introduction

Provide a broad background for Canada's productivity slowdown and give an overview of your analysis of TFP.

Part 1: Bank of Canada's report on productivity

Summarize the Senior Deputy Governor Carolyn Rogers's speech: "The Productivity Problem". In your summary, make sure to highlight the current challenges Canada is facing. Below are URLs of a video of the speech in addition to a text version. The video is divided into two parts: the speech itself and a Q&A session. The Q&A session provides additional detailed information, which could be useful in your write-up. Alternatively, you may read the text version of the same speech.

Video: <https://www.bankofcanada.ca/2024/03/productivity-problem/>

Text: <https://www.bankofcanada.ca/2024/03/time-to-break-the-glass-fixing-canadas-productivity-problem/>

Part 2: TFP analysis

Download the following two data series from FRED, Federal Reserve Bank of St. Louis (<https://fred.stlouisfed.org>) from 2001 to 2023 for Canada and your randomly assigned country (see the "List of randomly assigned country.xlsx" on Quercus). The frequency of all data series should be annual. To make sure you have the right data series, you should use the list of variable labels provided on Quercus (Data Label FRED.xlsx).¹

- Real GDP
- Population

Then, combine these two data series with the capital stock series provided on Quercus (see Capital Data Series.xlsx). Record the citation for each data series as a reference.

The objective for this section is to calculate the Solow Residual, a measure of the total factor productivity as follows. Starting from the following Cobb-Douglas production function:

$$Y = AK^{\alpha}N^{1-\alpha}$$

We can rearrange this equation for TFP and estimate the Solow Residual with:

$$A = \frac{Y}{K^{\alpha}N^{1-\alpha}}$$

Using this equation and the dataset, analyze the recent trends in TFP for Canada and compare them to the trends of your randomly assigned country. For these calculations, you will assume that $\alpha = 0.33$. Use the following steps:

1. Calculate TFP for Canada and your randomly assigned country.

¹ Hints: (1) You can search the code on the FRED website for the data (2) Use "Edit Graph" --> "Add Line" to add another data series on the same sheet

2. Create a measure of normalized TFP by dividing the series by the TFP in 2010 (i.e., the TFP in 2010 should be equal to 1).
 3. Create a single time series plot with the two countries' normalized TFP measures.
- Submit the spreadsheet on Quercus. As a best practice, the original data should not be altered!

Questions:

1. Compare the TFP in Canada to the TFP in your randomly assigned country, analyze and discuss whether Canada is having a slowdown during and after the COVID-19 period. Ensure you support your claims with evidence your plots.
2. Does the Canadian TFP series fit the description in the Senior Deputy Governor's speech? If not, what does the trend show instead?
3. If Canada's TFP is falling, what should the government do? What government interventions could increase TFP? Support your claims with evidence.

You must answer the above questions in a natural and connected manner. You may answer the questions in any order but ensure that each paragraph is related to the previous and is appropriately linked. Each paragraph should convey one major idea and have a logical internal flow that explains and defends that idea.

When you reference a data plot you must describe the variables it contains, and the trend shown by the data. Do not assume that the reader will look at the graphs and independently understand their meaning. You must guide your audience towards your conclusions within the body of the text.

Submission instructions

You must submit your assignment to **Quercus**. To do so, follow these steps:

- 1) Complete your essay in Microsoft Word.
- 2) Submit the Microsoft Word file and the spreadsheet file to the Quercus assignment.
- 3) **Must be completed before the deadline on Dec 4th at 11:59pm!**

You must verify your work was properly submitted before the deadline. You may make multiple submissions before the deadline if you discover an issue with a previous submission.

*Refer to the separate rubric for details on how your work will be evaluated.